QUINLAN A. CAMPBELL

quin@vt.edu | (919) 358-0889
www.linkedin.com/in/quinlancampbell
ORCID: https://orcid.org/0009-0003-0809-0725

EDUCATION

Virginia Tech **|** Bachelor of Science, May 2027
Major: Environmental Data Science **|** **Minor**: GIS **|** **Overall GPA**: 3.75
Relevant Coursework: Programming in Python, Intro to GIS, Modeling in GIS, Environmental Data Science, Systems Analysis, Remote Sensing, Forest Photogrammetry and Spatial Data Processing, Hydroinformatics, Biological Statistics

RESEARCH EXPERIENCE

AI, Drones, and Disaster Impact Research Assistant **|** Geography Department, Virginia Tech (May 2026–Present)
-Analyzed drone and aerial imagery to assess road damage and passability following hurricanes.
-Assisted in AI model validation and refinement by manually labeling image datasets and comparing human vs. machine judgments.
-Collaborated on data workflows using GIS and Python; contributed to manuscript preparation for publication.

Hydrochemistry Student Researcher **|** FREC Department, Virginia Tech (Aug. 2025-Present)
-Perform automated watershed delineation in R with LiDAR data and hydrology tools (terra, sf).
-Conduct terrain and slope analyses to classify strike- and dip-slope catchments and assess runoff.
-Support field sampling operations, including water chemistry collection and in situ sensor calibration.

Water Quality Lab Technician **|** Carey Lab, Virginia Tech (Mar. 2025-Present)
-Collect water quality samples and field observations across reservoir, inflow, and outflow sites.
-Support laboratory operations through field preparation, acid washing, data entry, and chemistry runs.
-Analyze water quality, macrophyte, and ecological monitoring data in R and ArcGIS.

Co-Lead Researcher- Iliamna corei **|** Restoration Ecology Lab, Virginia Tech (Aug. 2024-Present)
-Co-lead a research project focused on the restoration and monitoring of *Iliamna corei*, a rare plant.
-Conduct fieldwork including site visits, debris clearing, and habitat assessments to support restoration.
-Develop a long-term monitoring and restoration protocol to guide future conservation actions.
-Secure grant funding to implement conservation measures.

Soybean Laboratory Technician **|** Leisner Lab, Virginia Tech (Jan. 2025–May 2025)
-Conducted comprehensive plant care, including watering and managing soybean sample processing.
-Analyzed soybean samples to investigate the impact of elevated CO_2 on nutrient levels.
-Utilized advanced tools to measure transpiration and assess leaf physiology.

BIRDNET Research Assistant **|** Restoration Ecology Lab, Virginia Tech (Jan. 2025–May 2025)
-Processed and analyzed acoustic recordings using BirdNET to identify avian species presence across restored and unrestored stream sites.

Fuel Dynamics Field Research Assistant **|** FREC Department, Virginia Tech; Catoctin Mountain & Manassas Battlefield Park (May 2024–Aug. 2024)
-Assisted with field research across 340 plots in Catoctin Mountain Park and Manassas Battlefield Park.
-Conducted surveys on deadwood and trees, capturing DBH, canopy class, and decay class.
-Demonstrated strong plant identification skills while navigating diverse terrains.

ADDITIONAL PROFESSIONAL EXPERIENCE

***Disaster Divisions Intern* |** United Way Southwest Virginia (Jan. 2026-Present)
-Maintain and standardize disaster assistance case data, including household size, case status, and funding identifiers, to support recovery operations.
-Generate monthly and cumulative reports across multiple disaster grants and shared funding sources to track aid distribution and program status.
-Develop geospatial visualizations in ArcGIS to analyze spatial and temporal patterns in disaster assistance, including case clustering, funding intensity, and monthly distribution across service counties.
-Contribute to program evaluation and process improvement efforts related to disaster assistance and community recovery.

PUBLICATIONS

Campbell, Q., Sappestein, B., Hernandez, A., Willis, B., Fitzgerald, M., Lipscomb, M., Polk, N., Reid, R. E. B., Edwards, R., Klopf, R., & Reid, J. L. (in preparation). Population status and trend of Iliamna corei (Peter's Mountain mallow; Malvaceae), a federally endangered Appalachian endemic.

Reid, J. L., **Campbell, Q.,** Duerr, N., Fitzgerald, H. M., Judge, P., Polk, N. D., Ripa, G. N., Sapperstein, B., Staengl, E. J., & Vrh, S. (in review). Undammed: Freeing Rivers and Bringing Communities to Life [Book review]. Restoration Ecology.

Reid, J. L., Velo, M., Aparicio-Vera, S., **Campbell, Q.**, Colberg, E., Coscia, J. T., Fitzgerald, M., Frelich, L., Hyatt, L., & Logan, C. M. (2025). Tree planting choices mediate wildfire damage to tropical forest restoration in eastern Madagascar. Restoration Ecology. https://doi.org/10.1111/rec.70251

Reid, J. L., Burt, M. A., **Campbell, Q**., Coscia, J. T., Huston, J. C., Kubesch, J. O. C., MacDougal, P., Ripa, G. N., Staengl, E. J., & Viani, R. A. G. (2024). Ecological Restoration: Moving Forward Using Lessons Learned [Book review]. Restoration Ecology. https://doi.org/10.1111/rec.14163

PRESENTATIONS

Campbell, Q., Sapperstein, B., Reid, J.L., *Population Status and Trend of Iliamna corei (Peter's Mountain Mallow; Malvaceae), a Federally Endangered Appalachian Endemic.* Poster presentation, School of Plant and Environmental Science Symposium, Blacksburg, VA, 2026.

Campbell, Q., Sapperstein, B., Reid, J.L., *Population Status and Trend of Iliamna corei (Peter's Mountain Mallow; Malvaceae), a Federally Endangered Appalachian Endemic.* Poster presentation, Dennis Dean Undergraduate Research and Creative Scholarship Conference, Blacksburg, VA, 2026.

Campbell, Q., Gannon, J.P., *Hydrochemical Signatures of Scarp and Dip-Slope Drainage in the Valley and Ridge Region.* Poster presentation, Geological Society of America Joint Southeastern, North-Central, and South-Central Section Meeting, Memphis, TN, 2026.

Campbell, Q., Sapperstein, B., Reid, J.L., *Conservation and Management Procedures for Critically Endangered Peter's Mountain Mallow (Iliamna corei)*. Poster presentation, Biological and Ecological Restoration Initiative Symposium, Blacksburg, VA, 2025.

GRANTS, AWARDS, ACADEMIC HONORS

Ut Prosim Internship Support Fund (2026)
Global Change Center Undergraduate Research Grant (2024)
Dean's List (Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026)

LEADERSHIP, ACTIVITIES, SERVICE, VOLUNTEER WORK

***Service Chair, Philanthropy Chair, Pledge Parent* |** Lambda Iota Mu (Feb. 2024-Present)
Coordinating and leading service, philanthropy, and new-member initiatives through a conservation-focused service organization.

***Volunteer* |** Campus Kitchen (Nov. 2025-Present)
Coordinating the collection of surplus food from campus dining facilities and transporting it to local food banks to support community needs.

PROFESSIONAL MEMBERSHIPS AND AFFILIATIONS

Ecological Society of America (ESA)
American Geophysical Union (AGU)
Geological Society of America (GSA)

CERTIFICATIONS

FEMA IS-100.c – Introduction to the Incident Command System
FEMA IS-700.b – National Incident Management System (NIMS)

SKILLS

Data & Analysis: R, Python, ArcGIS Pro, GitHub, Excel, geospatial analysis, data visualization
Field & Laboratory Methods: Water quality sampling, plant identification, forest inventory, soil and leaf sampling, CN analysis, Li-COR gas exchange systems
Research & Computing: Reproducible workflows, data cleaning, GIS-based modeling, remote sensing interpretation